Week 1: Study Sheet

EE0849 Introduction to Robotics - Problems & Solutions

Table of contents

1 Vectors and Inner Products

1.1 Problem 1.1

Given vectors $\mathbf{a} = \begin{bmatrix} 3 \\ 4 \\ 0 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, compute:

a) The dot product $\mathbf{a} \cdot \mathbf{b}$
b) The magnitude of each vector
c) The angle between the vectors

1.1.1 Solution

a) Dot product:

$$\mathbf{a} \cdot \mathbf{b} = 3(1) + 4(0) + 0(2) = 3$$

b) Magnitudes:

$$\|\mathbf{a}\| = \sqrt{3^2 + 4^2 + 0^2} = \sqrt{25} = 5$$
$$\|\mathbf{b}\| = \sqrt{1^2 + 0^2 + 2^2} = \sqrt{5}$$

c) Angle:

$$\cos\theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|\|\mathbf{b}\|} = \frac{3}{5\sqrt{5}} = \frac{3}{5\sqrt{5}} \approx 0.268$$
$$\theta = \cos^{-1}(0.268) \approx 74.5°$$

1.2 Problem 1.2

Are vectors $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 4 \\ -1 \\ 2 \end{bmatrix}$ orthogonal?

1.2.1 Solution

Two vectors are orthogonal if $\mathbf{u} \cdot \mathbf{v} = 0$.

$$\mathbf{u} \cdot \mathbf{v} = 1(4) + 2(-1) + (-1)(2) = 4 - 2 - 2 = 0$$

Yes, the vectors are orthogonal.

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2 Cross Products and Skew-Symmetric Matrices

2.1 Problem 2.1

Given $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$ and $\mathbf{y} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$, compute $\mathbf{x} \times \mathbf{y}$.

2.1.1 Solution

Using the determinant formula:

$$\mathbf{x} \times \mathbf{y} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 1 & 3 \\ 1 & -1 & 2 \end{vmatrix}$$

$$= \mathbf{i}(1 \cdot 2 - 3 \cdot (-1)) - \mathbf{j}(2 \cdot 2 - 3 \cdot 1) + \mathbf{k}(2 \cdot (-1) - 1 \cdot 1)$$

$$= \mathbf{i}(2 + 3) - \mathbf{j}(4 - 3) + \mathbf{k}(-2 - 1)$$

$$\mathbf{x} \times \mathbf{y} = \begin{bmatrix} 5 \\ -1 \\ -3 \end{bmatrix}$$

2.2 Problem 2.2

Construct the skew-symmetric matrix $S(\mathbf{x})$ for $\mathbf{x} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$ and verify that $S(\mathbf{x})\mathbf{y} = \mathbf{x} \times \mathbf{y}$ for $\mathbf{y} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$.

2.2.1 Solution

The skew-symmetric matrix is:

$$S(\mathbf{x}) = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & -3 & 1 \\ 3 & 0 & -2 \\ -1 & 2 & 0 \end{bmatrix}$$

Computing $S(\mathbf{x})\mathbf{y}$:

$$S(\mathbf{x})\mathbf{y} = \begin{bmatrix} 0 & -3 & 1 \\ 3 & 0 & -2 \\ -1 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 + 3 + 2 \\ 3 + 0 - 4 \\ -1 - 2 + 0 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \\ -3 \end{bmatrix}$$

This matches $\mathbf{x} \times \mathbf{y}$ from Problem 2.1.

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3 Rotation Matrices

3.1 Problem 3.1

Verify that the following matrix is a valid rotation matrix:

$$R = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

3.1.1 Solution

A matrix R is a valid rotation matrix if:

1. $R^T R = I$ (orthogonality)
2. $\det(R) = 1$ (proper rotation)

Check orthogonality:

$$R^T R = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} + \frac{1}{2} & -\frac{1}{2} + \frac{1}{2} & 0 \\ -\frac{1}{2} + \frac{1}{2} & \frac{1}{2} + \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I \quad \checkmark$$

Check determinant:

$$\det(R) = 1 \cdot \left(\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} - \left(-\frac{1}{\sqrt{2}} \right) \cdot \frac{1}{\sqrt{2}} \right) = \frac{1}{2} + \frac{1}{2} = 1 \quad \checkmark$$

Yes, R is a valid rotation matrix (a 45° rotation about the z-axis).

3.2 Problem 3.2

Given a rotation matrix R, what is its inverse? Compute the inverse of:

$$R = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

3.2.1 Solution

For rotation matrices, the inverse equals the transpose: $R^{-1} = R^T$

$$R^{-1} = R^T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

Verification: $R \cdot R^T = I$

$$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \checkmark$$

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4 Elementary Rotations

4.1 Problem 4.1

Compute $R_z(30°)$ and use it to rotate the vector $\mathbf{v} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$.

4.1.1 Solution

With $\theta = 30°$: $\cos(30°) = \frac{\sqrt{3}}{2} \approx 0.866$, $\sin(30°) = \frac{1}{2} = 0.5$

$$R_z(30°) = \begin{bmatrix} \cos 30° & -\sin 30° & 0 \\ \sin 30° & \cos 30° & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Rotating the vector:

$$\mathbf{v}' = R_z(30°)\mathbf{v} = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0.866 \\ 0.5 \\ 0 \end{bmatrix}$$

The vector now points 30° from the x-axis in the xy-plane.

4.2 Problem 4.2

Compute $R_x(90°)$ and show what happens to the unit vectors $\hat{j} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ and $\hat{k} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$.

4.2.1 Solution

With $\theta = 90°$: $\cos(90°) = 0$, $\sin(90°) = 1$

$$R_x(90°) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos 90° & -\sin 90° \\ 0 & \sin 90° & \cos 90° \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

Rotating $\hat{j}$:

$$R_x(90°)\hat{j} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \hat{k}$$

Rotating $\hat{k}$:

$$R_x(90°)\hat{k} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} = -\hat{j}$$

The y-axis rotates to z, and z-axis rotates to negative y.

4.3 Problem 4.3

A robot arm first rotates 45° about z-axis, then 30° about the (new) x-axis. Find the total rotation matrix.

4.3.1 Solution

For successive rotations about the **current (body) frame axes**, we post-multiply:

$$R_{total} = R_z(45°) \cdot R_x(30°)$$

$$R_z(45°) = \begin{bmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} & 0 \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \approx \begin{bmatrix} 0.707 & -0.707 & 0 \\ 0.707 & 0.707 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_x(30°) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix} \approx \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.866 & -0.5 \\ 0 & 0.5 & 0.866 \end{bmatrix}$$

$$R_{total} = \begin{bmatrix} 0.707 & -0.707 & 0 \\ 0.707 & 0.707 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.866 & -0.5 \\ 0 & 0.5 & 0.866 \end{bmatrix}$$

$$= \begin{bmatrix} 0.707 & -0.612 & 0.354 \\ 0.707 & 0.612 & -0.354 \\ 0 & 0.5 & 0.866 \end{bmatrix}$$

5 Change of Frame

5.1 Problem 5.1

Frame $\{1\}$ is rotated 90° about the z-axis relative to frame $\{0\}$. A point has coordinates $\mathbf{p}^1 = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$ in frame $\{1\}$. Find its coordinates in frame $\{0\}$.

5.1.1 Solution

The rotation matrix from frame $\{1\}$ to frame $\{0\}$:

$$R_1^0 = R_z(90°) = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Transform coordinates:

$$\mathbf{p}^0 = R_1^0 \cdot \mathbf{p}^1 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$$

5.2 Problem 5.2

Frame $\{B\}$ is obtained by rotating frame $\{A\}$ by 60° about the x-axis. Vector $\mathbf{v}$ has coordinates $\mathbf{v}^A = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ in frame $\{A\}$. Find $\mathbf{v}^B$.

5.2.1 Solution

The rotation matrix from $\{A\}$ to $\{B\}$:

$$R_A^B = R_x(60°) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.5 & -0.866 \\ 0 & 0.866 & 0.5 \end{bmatrix}$$

To go from A coordinates to B coordinates, we use $R_B^A = (R_A^B)^T$:

$$\mathbf{v}^B = R_B^A \cdot \mathbf{v}^A = (R_A^B)^T \cdot \mathbf{v}^A$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.5 & 0.866 \\ 0 & -0.866 & 0.5 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.5 \\ -0.866 \end{bmatrix}$$

5.3 Problem 5.3 (Interpretation)

Explain the difference between:

a) Rotating a vector within a fixed frame
b) Expressing the same vector in a different frame

5.3.1 Solution

a) Rotation as Mapping (Active Transformation):

- The vector **physically moves** to a new position
- The frame stays fixed
- If $\mathbf{v}_{new} = R \cdot \mathbf{v}_{old}$, the vector rotates by the angle encoded in R
- Example: A robot arm rotating a held object

b) Change of Frame (Passive Transformation):

- The vector **stays fixed** in physical space
- We express its coordinates in a different reference frame
- If $\mathbf{v}^0 = R_1^0 \cdot \mathbf{v}^1$, we're converting coordinates from frame 1 to frame 0
- Example: The same point seen from two different cameras

Key insight: Mathematically similar operations, but physically different interpretations!

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6 Summary of Key Formulas

Concept	Formula
Dot product	$\mathbf{a} \cdot \mathbf{b} = \sum a_i b_i = \|\mathbf{a}\|\|\mathbf{b}\| \cos\theta$
Cross product	$\mathbf{a} \times \mathbf{b} = S(\mathbf{a})\mathbf{b}$
Skew-symmetric	$S(\mathbf{x}) = \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix}$
Rotation inverse	$R^{-1} = R^T$
Change of frame	$\mathbf{v}^0 = R_1^0 \cdot \mathbf{v}^1$
Rotation composition	$R_{total} = R_2 \cdot R_1$ (apply R_1 first)